



# PRECURSOR ANALYSIS REPORT: CONFICKER INFECTION OF GUNDREMMINGEN NUCLEAR POWER PLANT 2016

Cybersecurity for the Operational Technology  
Environment (CyOTE)

30 JUNE 2022



U.S. DEPARTMENT OF  
**ENERGY**

Office of  
Cybersecurity, Energy Security,  
and Emergency Response

This document was prepared by Idaho National Laboratory (INL) under an agreement with and funded by the U.S. Department of Energy.

INL is a U.S. Department of Energy National Laboratory operated by Battelle Energy Alliance, LLC

## TABLE OF CONTENTS

|                                                                                           |           |
|-------------------------------------------------------------------------------------------|-----------|
| <b>1. EXECUTIVE SUMMARY .....</b>                                                         | <b>2</b>  |
| <b>2. INTRODUCTION .....</b>                                                              | <b>3</b>  |
| 2.1. APPLYING THE CYOTE METHODOLOGY .....                                                 | 3         |
| 2.2. BACKGROUND ON THE ATTACK .....                                                       | 5         |
| <b>3. OBSERVABLE AND TECHNIQUE ANALYSIS .....</b>                                         | <b>8</b>  |
| 3.1. REPLICATION THROUGH REMOVABLE MEDIA TECHNIQUE (T0847) FOR INITIAL ACCESS .....       | 8         |
| 3.2. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR INITIAL ACCESS .....           | 9         |
| 3.3. REMOTE SERVICES TECHNIQUE (T0886) FOR INITIAL ACCESS .....                           | 10        |
| 3.4. NATIVE API TECHNIQUE (T0834) FOR EXECUTION .....                                     | 11        |
| 3.5. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION .....                                      | 12        |
| 3.6. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION .....                   | 13        |
| 3.7. ROOTKIT TECHNIQUE (T0851) FOR INHIBIT RESPONSE FUNCTION .....                        | 14        |
| 3.8. BLOCK REPORTING MESSAGE TECHNIQUE (T0804) FOR INHIBIT RESPONSE FUNCTION .....        | 16        |
| 3.9. ALARM SUPPRESSION TECHNIQUE (T0878) FOR INHIBIT RESPONSE FUNCTION .....              | 17        |
| 3.10. MASQUERADING TECHNIQUE (T0849) FOR EVASION .....                                    | 18        |
| 3.11. MODIFY PROGRAM TECHNIQUE (T0889) FOR PERSISTENCE .....                              | 19        |
| 3.12. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL ..... | 20        |
| 3.13. COMMONLY USED PORT TECHNIQUE (T0885) FOR COMMAND AND CONTROL .....                  | 21        |
| 3.14. REMOTE SYSTEM DISCOVERY TECHNIQUE (T0846) FOR DISCOVERY .....                       | 22        |
| 3.15. REMOTE SYSTEM INFORMATION DISCOVERY TECHNIQUE (T0888) FOR DISCOVERY .....           | 23        |
| 3.16. REMOTE SERVICES TECHNIQUE (T0886) FOR LATERAL MOVEMENT .....                        | 24        |
| 3.17. VALID ACCOUNTS TECHNIQUE (T0859) FOR LATERAL MOVEMENT .....                         | 25        |
| 3.18. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR LATERAL MOVEMENT .....        | 26        |
| 3.19. PROGRAM DOWNLOAD TECHNIQUE (T0843) FOR LATERAL MOVEMENT .....                       | 27        |
| 3.20. MANIPULATION OF CONTROL TECHNIQUE (T0831) FOR IMPACT .....                          | 28        |
| 3.21. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT .....                 | 29        |
| 3.22. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT .....                             | 30        |
| <b>APPENDIX A: OBSERVABLES LIBRARY .....</b>                                              | <b>31</b> |
| <b>APPENDIX B: ARTIFACTS LIBRARY .....</b>                                                | <b>35</b> |
| <b>APPENDIX C: OBSERVERS .....</b>                                                        | <b>47</b> |
| <b>REFERENCES.....</b>                                                                    | <b>48</b> |

## FIGURES

|                                           |          |
|-------------------------------------------|----------|
| <b>FIGURE 1. CYOTE METHODOLOGY .....</b>  | <b>3</b> |
| <b>FIGURE 2. INTRUSION TIMELINE .....</b> | <b>5</b> |

## TABLES

|                                                                         |          |
|-------------------------------------------------------------------------|----------|
| <b>TABLE 1. TECHNIQUES USED IN THE GUNDREMMINGEN CYBER-ATTACK .....</b> | <b>7</b> |
| <b>TABLE 2. PRECURSOR ANALYSIS REPORT QUANTITATIVE SUMMARY .....</b>    | <b>7</b> |

# PRECURSOR ANALYSIS: CONFICKER INFECTION OF GUNDREMMINGEN NUCLEAR POWER PLANT 2016

## 1. EXECUTIVE SUMMARY

The Conficker Infection of Gundremmingen Nuclear Power Plant 2016 Precursor Analysis Report leverages publicly available information about the global Conficker malware outbreak and catalogs anomalous observables for each technique employed in the attack against Gundremmingen. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

Conficker initially appeared in November 2008, and throughout its evolution has acquired several different names including Downup, Downadup, and Kido. Conficker is a worm-type malware that self-propagates across vulnerable systems. However, the Conficker Working Group, a coalition of cybersecurity researchers, was able to shut down all the domains associated with the Conficker command and control servers in 2009, eliminating the malware's extant ability to execute malicious code against infected systems.<sup>1</sup>

In the case of the Gundremmingen Nuclear Power Plant in Gundremmingen, Germany a routine security check on 24 April 2016 discovered Conficker malware on 18 systems within the plant's IT network that oversees the fuel handling system.<sup>2</sup> Upon detection of the malware, Gundremmingen shut down for two days to follow security procedures, including wiping the malware from all infected systems and scanning all other systems to ensure the malware was no longer present. Gundremmingen officials said they suspected someone brought in the malware by accident on a USB thumb drive, either from home or computers in the facility.<sup>3</sup> The plant's internal network was isolated from the Internet, obstructing the Conficker malware from further propagation to external machines or otherwise enabling adversary access to the infected internal machines.<sup>4</sup>

CyOTE researchers and analysts identified 22 techniques utilized during the attack, generating a total of 78 observables using MITRE's ATT&CK® for Industrial Control Systems framework. The CyOTE program assesses observables accompanying techniques used prior to the triggering event to identify opportunities to detect malicious activity. If observables accompanying the attack techniques are perceived and investigated prior to the triggering event, then earlier comprehension of malicious activity can take place. Eighteen of the identified techniques used during the Conficker cyber attack were precursors to the triggering event. Case study analysis identified 66 observables associated with these precursor techniques, 41 of which were assessed to have an increased likelihood of being perceived preceding the triggering event of the malware. The response and comprehension time likely could have been reduced by four hours if these highly perceivable observables had been identified before the triggering event occurred.

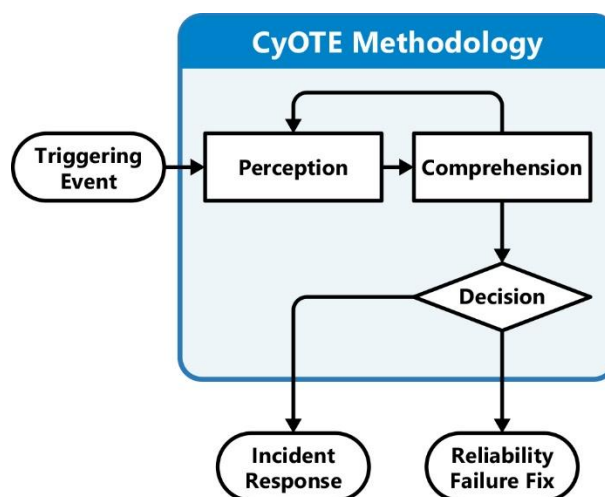
The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references to support the comprehension of indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

## 2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

### 2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in Figure 1. CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® for Industrial Control Systems (ICS) framework as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.



**Figure 1. CyOTE Methodology**

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the case study's cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references to support the comprehension of indicators of attack.

## 2.2. BACKGROUND ON THE ATTACK

The Microsoft Malware Protection Center identified the first iteration of Conficker malware (Conficker A) in November 2008. Conficker is a worm-type malware which infects unpatched Windows operating systems by utilizing three different methods of initial access to propagate and worm into connected systems. Conficker connects these machines to a remote server that allows the adversaries to execute remote code on infected systems, although this functionality was never used in a way that caused significant impact on victims. Conficker has at least five different variants (A through E), which researchers identified between November 2008 and April 2009. Information from researchers and victims indicated the different Conficker variants had infected 25 million machines as of 25 October 2011.<sup>5</sup> Despite software security responses, Conficker infections have continued to propagate in older systems years after the malware's introduction.

In the case presented in this study, a routine security check revealed a Conficker infection at the Gundremmingen Nuclear Power Plant on 24 April 2016.<sup>6</sup> Among the infected systems was a computer network containing data visualization software associated with the equipment the plant utilized for moving nuclear fuel rods.<sup>7</sup>

A timeline of adversarial techniques for a generic Conficker attack is shown in Figure 2. The timeline includes the estimated number of hours prior to and after the triggering event. The timeline after the triggering event includes the assessed victim comprehension timeline.

The Conficker malware replicates in three different ways: by removable media, network shares, or by exploitation of remote services. The malware at Gundremmingen was most likely introduced to the plant's machines by a USB storage device (H-4.5). When a Conficker-infected removable device connects to a machine and Conficker is installed, the malware sleeps for 30 minutes to decrease risk of detection (H-4).<sup>8</sup> In the case of Gundremmingen, the malware was unable to propagate beyond the plant's network or otherwise call back to adversary command and control (C2) servers because the plant's network was isolated from the Internet.<sup>9</sup>

After its sleep period, Conficker executes a multitude of evasive steps to avoid detection, including embedding itself in the form of a rootkit.<sup>10</sup> Immediately upon execution of the evasive steps, Conficker attempts to contact the adversary's domain to download any updates or further instructions. In the case of isolated networks such as Gundremmingen's, even though Conficker is unable to connect with the adversary's domain, it continues its execution process using peer-to-peer (P2P) user

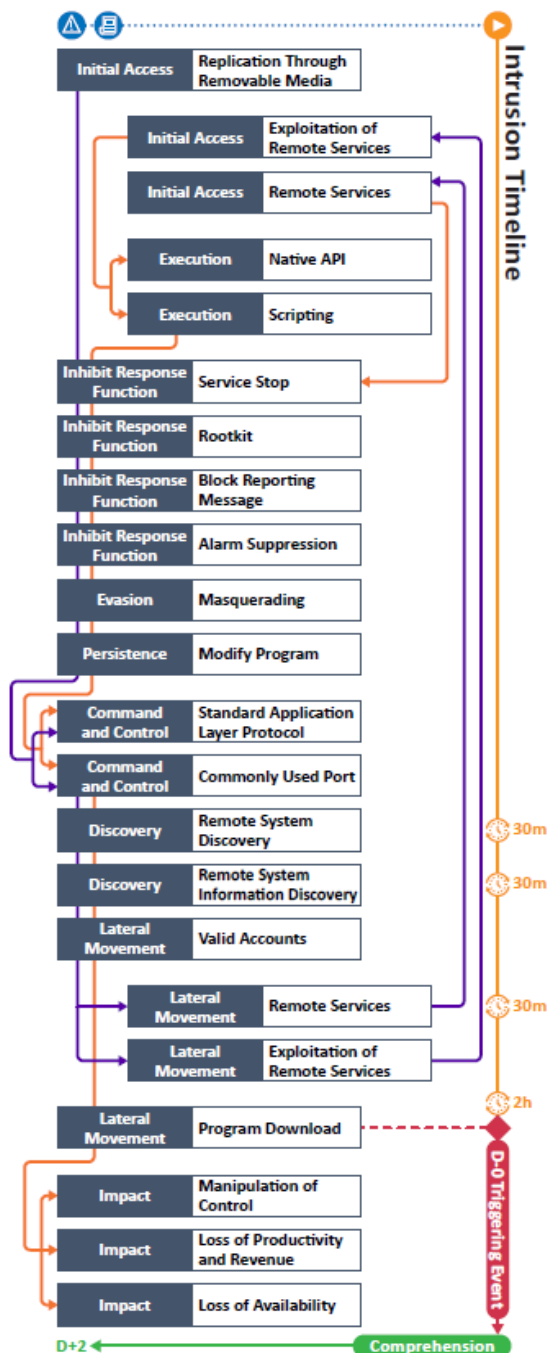


Figure 2. Intrusion Timeline

datagram protocol (UDP) scans to identify other machines on the network, after which Conficker again sleeps for 30 minutes (H-3).<sup>11</sup>

Next, Conficker uses an HTTP server created earlier as part of the adversary's domain connection process to communicate with machines found in the P2P scan and determine if those machines are already infected.<sup>12</sup> Once this process is completed, Conficker sleeps for another 30 minutes (H-2.5) before attempting to infect any uninfected machines by utilizing network shares to replicate. The malware then sleeps for a fourth time for 30 minutes (H-2) before attempting to infect any remaining uninfected machines through vulnerability exploitation and brute-force.<sup>13,14</sup>

Finally, the malware installs W.32 Waledac and SpywareProtect2009 and sleeps for two hours before executing any further impact steps, including execution of arbitrary code from a connected C2 server, toward a triggering event (D-0).<sup>15</sup>

Upon detection of the malware at Gundremmingen, personnel shut down operations for two days (D+2) to wipe the malware from infected systems and scan other systems to ensure they were clean.

Conficker was never utilized by threat actors to execute any further impact steps, despite being available for remote code execution. The malware was thwarted by a group that shut down all the domains associated with the C2 servers, leaving Conficker impotent beyond its inherent propagation capability. The Conficker malware could have caused further impact to Internet-connected machines assuming adversaries were able to leverage its capabilities.

Analysis identified 22 techniques in a sequence and timeframe likely used by adversaries during this cyber attack (Table 1). These attack techniques are defined according to MITRE's ATT&CK® for ICS framework.

**Table 1. Techniques Used in the Gundremmingen Cyber-Attack**

| Initial Access                      | Execution                 | Persistence            | Privilege Escalation                  | Evasion                   | Discovery                           | Lateral Movement                | Collection                         | Command and Control                 | Inhibit Response Function     | Impair Process Control       | Impact                           |
|-------------------------------------|---------------------------|------------------------|---------------------------------------|---------------------------|-------------------------------------|---------------------------------|------------------------------------|-------------------------------------|-------------------------------|------------------------------|----------------------------------|
| Data Historian Compromise           | Change Operating Mode     | Modify Program         | Exploitation for Privilege Escalation | Change Operating Mode     | Network Connection Enumeration      | Default Credentials             | Automated Collection               | Commonly Used Port                  | Activate Firmware Update Mode | Brute Force I/O              | Damage to Property               |
| Drive-by Compromise                 | Command-Line Interface    | Module Firmware        | Hooking                               | Exploitation for Evasion  | Network Sniffing                    | Exploitation of Remote Services | Data from Information Repositories | Connection Proxy                    | Alarm Suppression             | Modify Parameter             | Denial of Control                |
| Engineering Workstation Compromise  | Execution through API     | Project File Infection |                                       | Indicator Removal on Host | Remote System Discovery             | Lateral Tool Transfer           | Detect Operating Mode              | Standard Application Layer Protocol | Block Command Message         | Module Firmware              | Denial of View                   |
| Exploit Public-Facing Application   | Graphical User Interface  | System Firmware        |                                       | Masquerading              | Remote System Information Discovery | Program Download                | I/O Image                          |                                     | Block Reporting Message       | Spoof Reporting Message      | Loss of Availability             |
| Exploitation of Remote Services     | Hooking                   | Valid Accounts         |                                       | Rootkit                   | Wireless Sniffing                   | Remote Services                 | Man in the Middle                  |                                     | Block Serial COM              | Unauthorized Command Message | Loss of Control                  |
| External Remote Services            | Modify Controller Tasking |                        |                                       | Spoof Reporting Message   |                                     | Valid Accounts                  | Monitor Process State              |                                     | Data Destruction              |                              | Loss of Productivity and Revenue |
| Internet Accessible Device          | Native API                |                        |                                       |                           |                                     |                                 | Point & Tag Identification         |                                     | Denial of Service             |                              | Loss of Protection               |
| Remote Services                     | Scripting                 |                        |                                       |                           |                                     |                                 | Program Upload                     |                                     | Device Restart/Shutdown       |                              | Loss of Safety                   |
| Replication Through Removable Media | User Execution            |                        |                                       |                           |                                     |                                 | Screen Capture                     |                                     | Manipulate I/O Image          |                              | Loss of View                     |
| Rogue Master                        |                           |                        |                                       |                           |                                     |                                 | Wireless Sniffing                  |                                     | Modify Alarm Settings         |                              | Manipulation of Control          |
| Spearphishing Attachment            |                           |                        |                                       |                           |                                     |                                 |                                    |                                     | Rootkit                       |                              | Manipulation of View             |
| Supply Chain Compromise             |                           |                        |                                       |                           |                                     |                                 |                                    |                                     | Service Stop                  |                              | Theft of Operational Information |
| Wireless Compromise                 |                           |                        |                                       |                           |                                     |                                 |                                    |                                     | System Firmware               |                              |                                  |

**Table 2. Precursor Analysis Report Quantitative Summary**

| Precursor Analysis Report Quantitative Summary     | Totals |
|----------------------------------------------------|--------|
| MITRE ATT&CK® for ICS Techniques                   | 22     |
| Technique Observables                              | 78     |
| Precursor Techniques                               | 18     |
| Precursor Technique Observables                    | 66     |
| Highly Perceivable Precursor Technique Observables | 41     |

### 3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

#### 3.1. REPLICATION THROUGH REMOVABLE MEDIA TECHNIQUE (T0847) FOR INITIAL ACCESS

The Conficker malware uses the Replication Through Removable Media technique (T0847) by leveraging the default autorun feature (autorun.inf) of Windows.<sup>16</sup> The autorun.inf function enables Conficker to automatically run executables whenever a removable drive is inserted into a system.<sup>17</sup> To increase its chance of success, the malware sets the “shell execute” prompt in the autorun.inf file to be “Open folder to view files”. The prompt is created with social engineering tactics in mind to interact with users unaware of the malware executing the autorun program, furthering its propagation across the network. Once the Conficker malware has its initial foothold, it listens for the WM\_DEVICECHANGE message. The WM\_DEVICECHANGE signals a new removable device or drive has been mapped to the compromised system and initiates an immediate attempt to infect the new device.<sup>18</sup>

Observers of this technique would include all computer users being requested to initiate the propagation (OT Cybersecurity, IT Cybersecurity, and IT Staff). Observers would likely see a notification in the Windows Event Log when a Universal Serial Bus (USB) or other removeable media device connects to the system. The newly attached device would automatically assign a drive signature.

A total of six observables were identified with the use of the [Replication Through Removeable Media technique \(T0847\)](#). This technique is important for investigation due to its autorun feature and its direct execution by a user. This technique modifies the host operating system files via the installation of the Conficker Malware, resulting in the host being commanded into a modified / compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired. This technique is used early in the attack timeline and responding to it will disrupt further infection of a network. Terminating the chain of techniques at this point would limit operational impact to an attempted malware infection.

Of the six observables associated with this technique, five are assessed to be highly perceivable (Drive Creation; File Creation; autorun.inf Execution Feature; Autoplay Alert, Windows Event Log entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                                 |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 48 artifacts could be generated by the Replication Through Removable Media technique |
| <b>Technique Observers<sup>a</sup></b>                        | IT Cybersecurity, IT Staff, OT Cybersecurity                                                    |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                                  |

<sup>a</sup> Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

### 3.2. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR INITIAL ACCESS

Once Conficker has gained initial access, the malware uses the Remote Procedure Call (RPC) communication protocol as its primary access point for propagation. This exploit utilizes the Microsoft Windows Server Service RPC Handling Remote Code Execution (RCE) Vulnerability ([\(BID 31874\)](#)), which Microsoft announced in October of 2008 and has since patched ([\(MS08-067\)](#)).<sup>19</sup> This vulnerability makes legacy Windows systems (e.g., Windows 2000, XP, Server 2003, Vista, and Server 2008) viable targets. Conficker exposes the RPC Vulnerability and targets TCP Port 445 using the Server Message Block (SMB) network file sharing protocol to execute arbitrary code without prior authentication. The malware proceeds to create an HTTP server on a random port. It exploits shared drive connections to other remote systems, directing them to connect and download the Conficker malware, further spreading the infection.<sup>20</sup>

Observers of this technique would include IT Staff, IT Cybersecurity and OT Cybersecurity with varying levels of awareness. Cybersecurity and IT Staff would require a network baseline in order to observe a new locally sourced HTTP server within the domain. OT Cybersecurity may notice latency and an increase in TCP 445 traffic within Windows Event Logs.

A total of five observables were identified with the use of [the Exploitation of Remote Services technique \(T0866\)](#). This technique is important for investigation due to its main vector of infection through a patched vulnerability ([\(MS08-067\)](#)). This technique appears early in the attack chain timeline and responding to it will prevent further damaging effects present in the malware itself. Terminating the chain of techniques at this point would prevent further intrusion.

Of the five observables associated with this technique, three are assessed to be highly perceivable (RPC Traffic; TCP 445 (SMB) Network File; Windows Event Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                             |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Artifacts                                                     | A total of 31 artifacts could be generated by the Exploitation of Remote Services technique |
| Technique Observers                                           | OT Cybersecurity, IT Cybersecurity, IT Staff                                                |
| Resources                                                     | <a href="#">Technique Detection References</a>                                              |

### 3.3. REMOTE SERVICES TECHNIQUE (T0886) FOR INITIAL ACCESS

Conficker's persistence and ability to infect systems relies on multiple pathways to continually spread into system networks. The malware does this by using the Remote Services technique (T0886) to attempt to copy additional instances to other machines through the administrative network share (ADMIN\$). The NetServerEnum request returns all visible Windows machines found on the network and the malware attempts to infect each of the identified machines.<sup>21</sup>

Conficker then harvests locally logged-on user credentials, copying itself to network connected machines. If the user credentials prove ineffective, Conficker will attempt authentication with all user names found on the local machine, leaving as an observable Windows Event ID 4624: Login, attempting brute-force authentication using nearly 250 common passwords.<sup>22</sup> Login failure will be reflected in Event ID 4625: Login Failures, and lead to user accounts being locked from too many invalid login attempts if standard security policy rules are applied.

The ability to observe this technique depends on the number of attempts made to breach user accounts and if locally logged-on user credentials are valid. If Conficker successfully uses harvested credentials to provide access across the ADMIN\$ drive, then users have little chance of observing the technique. Conficker's brute-force of username and password feature would cause users to be locked out after too many invalid login attempts. OT Cybersecurity, IT Cybersecurity, or IT Staff would be notified of login disruption by security alert policies focusing on Event ID 4625: Login Failures.

A total of four observables were identified with the use of the [Remote Services technique \(T0886\)](#). This technique is important for investigation as access to various accounts allows adversaries to move between assets and network segments, resulting in the ability to configure systems. This technique appears early on the attack timeline and terminating the chain of techniques at this point would limit the adversaries to initial access points, resulting in a halt of attack execution. Additionally, terminating the chain of techniques at this point would likely prevent the malware from spreading to other systems.

Of the four observables associated with this technique, two are assessed to be highly perceivable (Windows Event ID 4624; Windows Event ID 4625).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                             |
|---------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 24 artifacts could be generated by the Remote Services technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                              |

### 3.4. NATIVE API TECHNIQUE (T0834) FOR EXECUTION

Conficker disables access to vulnerabilities within the netapi32.dll and prevents access to this vector by other adversaries. The malware provides an in-memory patch to the RPC vulnerability ([MS08-067](#)) within the netapi32.dll, which removes the buffer overflow vector used to compromise the system. However, while the Conficker patch of the buffer overflow vector does protect against other adversaries, the malware authors specifically crafted the patch to allow other Conficker hosts to reinfect the victim. Conficker parses incoming RPC requests to search for a specific string within the Conficker shellcode, allowing remote access to victims when adversaries encounter the string.

A total of six observables were identified with the use of the [Native API technique \(T0834\)](#). This technique is important for investigation because it presents noticeable effects, such as the generation of anomalous HTTP traffic, anomalous DNS requests, and the insertion of new domains into the environment. This technique appears early in the timeline and responding to it will effectively halt all future events, including modification of host OS files and access to additional Conficker-infected hosts. This technique modifies the host operating system files, leaving the host in a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

OT Cybersecurity, IT Cybersecurity, or IT Staff would perceive these network-based observables if a network baseline is established and network traffic security protocols actively monitor outbound/inbound traffic.

Of the six observables associated with this technique, five are assessed to be highly perceivable (RPC Logs Potentially in Windows Event Logs; Visible on Network if SMB is not Encrypted; Connection to external IPs via HTTP; DNS Request Associated with Bad Domain; New Domain Inserted into the Environment).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                        |
|---------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 25 artifacts could be generated by the Native API technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                           |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                         |

### 3.5. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

After gaining initial access, Conficker further propagates by initiating a SMB session on Port 445/TCP with the targets discovered by the NetServerEnum execution. The infected host binds to the SRVSVC pipe and sends a NetPathCanonicalize request embedded with the exploit payload. The exploit prompts the target victim to contact the infected host on a connect-back port to download a portable executable (PE) DLL file. The exploit issues a Windows API call, so the service executes as a svchost.exe.<sup>23</sup>

This technique would require a network baseline to observe, limiting the number of observers who would be able to perceive the malware within their network. Due to its complexity, the observers, including IT Staff, IT Cybersecurity, and OT Cybersecurity, would have limited visibility.

A total of one observable was identified with the use of the [Scripting technique \(T0853\)](#). This technique is important for investigation due to its ability to assist with the propagation of the malware across networks. This technique appears after initial access is achieved and assists with access to other connected devices. Responding to it will effectively halt an infection vector used by the malware. Terminating the chain of techniques at this point would result in minimal impact within an environment and stop the proliferation of the malware across similarly connected devices.

None of the observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                       |
|---------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 12 artifacts could be generated by the Scripting technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                          |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                        |

### 3.6. SERVICE STOP TECHNIQUE (T0881) FOR INHIBIT RESPONSE FUNCTION

To limit response capabilities, the malware worm monitors domain name system (DNS) requests to domains containing anti-Conficker strings, antivirus resources, and malware support. When an applicable DNS is found, user access appears as if the network request timed out and DNS is retransmitted, resulting in a blocked site.<sup>24</sup> This technique inhibits global census or distributing new binary updates to infected systems.<sup>25</sup>

OT Cybersecurity, IT Cybersecurity, and IT Staff may observe backups, deletion of restore points, and disabling of automatic backup settings.

A total of five observables were identified with the use of the [Service Stop technique \(T0881\)](#). This technique is important for investigation because malware worm authors typically generate new variants (e.g., Conficker A, Conficker B, Conficker C). This technique appears early in the timeline and responding to it allows infected machines to have access to sites to update security software. Terminating the chain of techniques at this point would limit the widespread effects of the malware worm. This technique modifies the host operating system files, via the modification of backup images and restore points, resulting in the host placing the device into a modified or compromised state. If system backups are created after this technique is executed, then data recovery and disaster recovery efforts will be impaired. Additionally, due to the deletion of backup images and restore points, this technique impairs data recovery and disaster recovery efforts.

Of the five observables associated with this technique, four are assessed to be highly perceivable (DNS Request Time Out; Increase in DNS Retransmissions; Disables Automatic Backup Settings; Deletes Backup/Restore Points).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                          |
|---------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 13 artifacts could be generated by the Service Stop technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                             |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                           |

### 3.7. ROOTKIT TECHNIQUE (T0851) FOR INHIBIT RESPONSE FUNCTION

To limit response capabilities, Conficker implements a rootkit. This step ensures connections to remote security sites are prevented and appear to users to time out in order to evade security software.<sup>26</sup> The worm's embedded rootkit blocks access to domain requests containing any of the strings depicted [here](#). After the DNS retransmissions, the DNS request times out, creating the appearance of a connection failure.<sup>27</sup>

This increase in DNS retransmissions would show a slight increase in network latency. IT Staff, IT Cybersecurity, and OT Cybersecurity would receive alerts built on a network baseline if one had been established.

|             |                |             |                   |                  |                    |
|-------------|----------------|-------------|-------------------|------------------|--------------------|
| Activescan  | Adware         | Agnitum     | Ahnlab            | Anti-            | Antivir            |
| Arcabit     | Avast          | Avg         | Avgate            | Avrira           | Avp.               |
| Ac-sc       | Bdtools        | Bit9        | Bothunter         | Ca               | Castlecops         |
| Ccollomb    | Centralcommand | Cert        | Clamav            | Comodo           | Computerassociates |
| Confick     | Coresecure     | Cpsecure    | Cyber-ta          | Defender         | Downad             |
| Doxpara     | Drweb          | Dslreports  | Emsisoft          | Enigma           | Esafe              |
| Eset        | Etrust         | Ewido       | Fortinet          | F-prot           | Freeav             |
| Free-av     | Fsecure        | F-secure    | Gdata             | Gmer             | Grisoft            |
| Hackerwatch | Hackersoft     | Hauri       | Honey             | Ikarus           | Insecure           |
| iv.cs.uni   | Jotti          | K7computing | Kaspersky         | Kav              | Kido               |
| Llnw        | Llnwd.         | Malware     | Mcafee            | Microsoft        | Mirage             |
| Mitre       | Msdn.          | Msft        | Msftncs           | Ms-mvp           | Msmvps             |
| Mtc.sri     | Nai            | Ncircle     | Networkassociates | Nmap             | Nod32              |
| Norman      | Norton         | Onecare     | Panda             | Pctools          | Precisesecurity    |
| Prevx       | Ptsecurity     | Qualys      | Quickheal         | Removal          | Rising             |
| Rootkit     | Safety.live    | Sans        | Secunia           | Secure computing | Secureworks        |
| Snort       | Sophos         | Spamhaus    | Spyware           | Staysafe         | Sunbelt            |
| Symantec    | Technet        | Tenablese   | Threat            | Threatexpert     | Trendmicro         |
| Trojan      | Vet            | Virscan     | Virus             | Wilderssecurity  | windowsupdate      |

A total of three observables were identified with the use of the [Rootkit technique \(T0851\)](#). This technique is important for investigation due to its noticeable user account lockdown effects and password security alerts, increasing awareness from the average user to the IT Department. This technique appears toward the end of the malware's infection phase and responding to it will alert IT of the malware's presence inside a network. Terminating the chain of techniques at this point would increase the effectiveness of actions taken to limit further spreading within a given network.

This technique modifies the host operating system files via the implementation of the rootkit, resulting in rootkit executables placing the host into a modified or compromised state. If system backups are created after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the three observables associated with this technique, all three are assessed to be highly perceivable (Network Request Timeouts from Security Related Websites; Increase in DNS Retransmissions; Blocking of Predetermined DNS Requests).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                     |
|---------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 23 artifacts could be generated by the Rootkit technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                        |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                      |

### 3.8. BLOCK REPORTING MESSAGE TECHNIQUE (T0804) FOR INHIBIT RESPONSE FUNCTION

To further disguise its activities, Conficker disables Window Security notifications, preventing the system from alerting the user that the malware is modifying registries. Access to SafeMode is prevented in the malware modified windows environment.<sup>28</sup>

Observers likely to see anomalous entries in the Event Viewer Log include OT Cybersecurity, IT Cybersecurity, and IT Staff.

A total of three observables were identified with the use of the [Block Reporting Message technique \(T0804\)](#). This technique is important for investigation because adversaries blocking report messages from the end user can lead to disabling a response function. This technique appears further along in the timeline and responding to it allows users to observe the current state of equipment. Terminating the chain of techniques at this point would allow users to respond in a proper or timely manner to an event, such as a dangerous fault. This technique modifies the host operating system files via the manipulation of system configuration settings and associated system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the three observables associated with this technique, two are assessed to be highly perceivable (Prevents Access to Safe Mode; Event Viewer Log Entry).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                     |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 18 artifacts could be generated by the Block Reporting Message technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                        |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                      |

### 3.9. ALARM SUPPRESSION TECHNIQUE (T0878) FOR INHIBIT RESPONSE FUNCTION

As an additional attempt to obscure evidence, Conficker modifies registries, disabling most mainstream security software programs from notifying staff of subsequent events. This step also prevents access to Safe Mode and disables Windows Security Alert notifications.<sup>29</sup>

OT Cybersecurity, IT Cybersecurity, and IT Staff may have observed modification within Event Viewer Log entries.

A total of three observables are identified with the use of the [Alarm Suppression technique \(T0878\)](#). This technique is important for investigation because it suppresses alerts built into security systems intended to alert staff of potential risks. This technique appears early in the timeline, and responding to it will limit operational impact, as staff would be aware of anomalous behavior in a system, allowing for earlier response. This technique modifies the host operating system files via the manipulation of system configuration settings and associated system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the three observables associated with this technique, two are assessed to be highly perceivable (Disabling of Security Software; Event Viewer Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                               |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 13 artifacts could be generated by the Alarm Suppression technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                  |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                |

### 3.10. MASQUERADING TECHNIQUE (T0849) FOR EVASION

Conficker uses a simple but effective mechanism to cloak its runtime presence. Although the service starts through svchost.exe, it is not visible in the service manager because its DisplayName is set to be empty and type is set to be invisible. Then, unlike well-behaved DLLs, the Conficker DLL initialization function never returns, hence it is not added to the DLL list of the given process. Since the DLL is added as part of a group that includes other normal services in the netsvcs group, the instance of svchost.exe is not terminated, allowing Conficker to run behind the scenes. Another masquerading capability of Conficker is to remove an offensive registry key, reboot the system, and delete the corresponding DLL file from the system32 directory.<sup>30</sup>

OT Cybersecurity, IT Cybersecurity, and IT Staff could observe DLL creation, renaming, and deletion.

A total of two observables were identified with the use of the [Masquerading technique \(T0849\)](#). This technique is important for investigation as adversaries purposely disguise files so staff may not suspect malicious applications or executables. This technique appears further along the attack timeline and responding to it would allow staff to stop execution of malicious files. Terminating the chain of techniques at this point would limit adversaries to reconnaissance activity. This technique modifies the host operating system files via the manipulation of system configuration settings and associated system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the two observables associated with this technique, one is assessed to be highly perceivable (Registered DLLs with DisplayName Set to Empty and Type Set to Invisible).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                          |
|---------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 15 artifacts could be generated by the Masquerading technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                             |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                           |

### 3.11. MODIFY PROGRAM TECHNIQUE (T0889) FOR PERSISTENCE

After the malware has ensured it will remain undetected on the system, it connects to one of its predetermined external IP address domains to see if it needs to update itself. If it does, then the malware will connect and a HTTP server is created on the infected machine. This HTTP server is leveraged later for lateral movement. Meanwhile, as the files download, the Conficker malware updates to the most recent version.<sup>31</sup>

OT Cybersecurity, IT Cybersecurity, and IT Staff may observe anomalous entries within an Event Viewer Log.

A total of four observables were identified with the use of the [Modify Program technique \(T0889\)](#). This technique is important for investigation as modifying programs and ensuring proper updates are installed affects how the malware interacts with the different systems. This technique appears further along in the timeline and responding to it will prevent malware updates from occurring. This technique modifies the host operating system files via creation of the HTTP server and modification of system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the four observables associated with this technique, two are assessed to be highly perceivable (Download of Files; Event Viewer Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                               |
|---------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of three artifacts could be generated by the Modify Program technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                  |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                |

### 3.12. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND AND CONTROL

Conficker queries a list of random domains, checking for ones that are live. The malware then attempts to resolve each generated domain name to an IP address. If it succeeds, it sends an HTTP request to the resolved domain. Data from the HTTP request is parsed and saved to memory. Conficker verifies the signature of any data received from this HTTP request, determining if the bytes returned are authentic and signed. If the signature is valid, then the data received is decrypted and executed. This step updates Conficker using only “authentic” sources, as determined by the signature of the returned bytes.

OT Cybersecurity, IT Cybersecurity, or IT Staff would perceive these network-based observables if a network baseline is established and network traffic security protocols actively monitor outbound/inbound traffic. The unauthorized outbound HTTP request would alert Cybersecurity personnel and IT Staff that anomalous activity was occurring in the network, prompting a cyber incident investigation.

A total of five observables were identified with the use of the [Standard Application Layer Protocol technique \(T0869\)](#). This technique is important for investigation due to its outbound connection and development of adversary-generated domains within a network. This technique appears before the proliferation stage and responding to it will limit the outbound traffic and connection to malicious URL domains. Terminating the chain of techniques at this point would stop the connection to malicious domains and therefore not allow for further adversary activity. This technique potentially modifies the host operating system files via the potential download and installation of signed data, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the five observables associated with this technique, three are assessed to be highly perceivable (Targeting TCP 445 (SMB); Connection to Malicious URL Domains for C2C; Windows Event Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                                 |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 12 artifacts could be generated by the Standard Application Layer Protocol technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                                    |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                                  |

### 3.13. COMMONLY USED PORT TECHNIQUE (T0885) FOR COMMAND AND CONTROL

The Conficker malware abuses the RPC vulnerability ([MS08-067](#)) through the use of the commonly used Port 445/TCP to initiate an SMB session. The malware can then bind to the SRVSVC pipe and issue the NetPathCanonicalize request, which houses the exploit payload.<sup>32</sup>

Windows Event Log entries indicate the creation of an HTTP Server and an increase in RPC traffic would be observable to those with an established baseline. Observers of this technique would primarily be those associated with the monitoring of network traffic and endpoint monitoring, namely through the anomalous RPC and SMB traffic, as well as Windows Log entries and HTTP server creation. Observers of this technique would include IT staff, IT Cybersecurity and OT Cybersecurity personnel.

A total of five observables were identified with the use of the [Commonly Used Port technique \(T0885\)](#). This technique is important for investigation as adversaries may communicate over a commonly used port to bypass firewalls or network detection systems. This technique appears relatively late in the timeline and may alert personnel to abnormal activity, prompting a cyber incident response. Terminating the chain of techniques at this point would allow personnel to investigate unusual network traffic, decreasing comprehension time.

Of the five observables associated with this technique, three are assessed to be highly perceivable (RPC Traffic; Targeting TCP 445 (SMB); Windows Event Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                  |
|---------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of five artifacts could be generated by the Commonly Used Port technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                     |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                   |

### 3.14. REMOTE SYSTEM DISCOVERY TECHNIQUE (T0846) FOR DISCOVERY

Conficker utilizes the HTTP server created in the [Modify Program technique \(T0889\)](#) to perform peer-to-peer (P2P) UDP scans. These scans are conducted on Port 53/UDP to identify other machines on the network by targeting random IP addresses. This scanning is the initial P2P setup routine for W32.Downadup.C.<sup>33</sup>

If traffic monitoring is executed properly, OT Cybersecurity, IT Cybersecurity, and IT Staff should observe an increase in UDP traffic compared to baseline activity.

A total of two observables were identified with the use of the [Remote System Discovery technique \(T0846\)](#). This technique is important for investigation as discovering a list of other systems on a network by IP address, hostname, or other logical identifiers could allow adversaries to laterally move throughout a network. This technique appears late in the timeline, close to the triggering event, and responding to it will effectively prevent the spread of Conficker throughout machines on a network.

Of the two observables associated with this technique, one is assessed to be highly perceivable (High UDP Traffic).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                     |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 43 artifacts could be generated by the Remote System Discovery technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                        |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                      |

### 3.15. REMOTE SYSTEM INFORMATION DISCOVERY TECHNIQUE (T0888) FOR DISCOVERY

Conficker uses the Remote System Information Discovery technique (T0888) in multiple ways. The malware includes a P2P mechanism, enabling machines infected with Conficker variants C through E to communicate, instructing them to download W32.Waledac.<sup>34</sup> Conficker can act as both a client and a server to share content, both distributing to and receiving from other Conficker-infected computers. Conficker includes an algorithm to deterministically generate a set of domain names each day and contact those domains throughout the day to check if an updated version of the worm is available for download.

Four scanning threads generate a random list of IP addresses for the hosts to contact. The same IP-to-port mapping is used to help determine which ports on which IPs could be targeted. Conficker then uses an HTTP server to communicate with another device and determine whether it is already infected. A Conficker host can have up to 32 Conficker server instances running, with each IP address generating a server instance. Conficker simultaneously checks the binary updates of the infected computers to see if the Conficker-only content is digitally signed by Conficker authors to distribute between peers. Terminating the chain of techniques at this point would likely prevent the malware from spreading to other systems.

Observers of this technique may include OT Cybersecurity, IT Cybersecurity, and IT Staff.

There were three observables identified with the use of [the Remote System Information Discovery technique \(T0888\)](#). This technique is important for investigation since adversaries may attempt to get a listing of other systems by IP address, hostname, or other logical identifiers on a network that is used for lateral movement or discovery techniques.

Of the three observables associated with this technique, one is assessed to be highly perceivable (HTTP request (getmyip.org, getmyip.co.uk, checkip.dyndns.org)).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                                    |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Artifacts                                                     | A total of eight artifacts could be generated by the Remote System Information Discovery technique |
| Technique Observers                                           | OT Cybersecurity, IT Cybersecurity, IT Staff                                                       |
| Resources                                                     | <a href="#">Technique Detection References</a>                                                     |

### 3.16. REMOTE SERVICES TECHNIQUE (T0886) FOR LATERAL MOVEMENT

The Remote Services technique (T0886) occurs as Conficker uses network shares to further spread itself throughout a given network. In doing this, Conficker runs rundll32.exe to copy itself through the system and will sleep after this technique is performed to prevent itself from being detected. It also identifies admin share drives or Inter-process Communication (IPC).<sup>35</sup>

OT Cybersecurity, IT Cybersecurity, and IT Staff may observe the use of this technique.

A total of two observables were identified with the use of the [Remote Services technique \(T0886\)](#). This technique is important for investigation as it allows adversaries to move between assets and network segments, resulting in the ability to configure systems. This technique appears both early on the attack timeline with initial access while later being used for lateral movement into another systems. Terminating the chain of techniques at this point would limit the malware's spread throughout the network.

Of the two observables associated with this technique, none are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                             |
|---------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 24 artifacts could be generated by the Remote Services technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                              |

### 3.17. VALID ACCOUNTS TECHNIQUE (T0859) FOR LATERAL MOVEMENT

The next step in Conficker's attack chain is to query valid user accounts on a target remote server. To authenticate with the target, Conficker tries the credentials of the locally logged-on user first. If these credentials do not work, then the malware begins trying different username and password pairs.<sup>36</sup> The malware attempts to brute-force these accounts by using a table of 240 common passwords.<sup>37</sup>

With security policy best practices implementation, OT Cybersecurity, IT Cybersecurity, and IT Staff could observe network traffic associated with this activity, such as sending password lists, querying users, and communication between targets.

A total of two observables were identified with the use of the [Valid Accounts technique \(T0859\)](#). This technique is important for investigation because compromised credentials are used to bypass access controls to various resources within a network or grant an adversary increased privileges to specific systems and devices. This technique appears later in the timeline for this case and responding to it will effectively limit access to protected machines. Terminating the chain of techniques at this point would likely prevent the malware from spreading to other systems.

Of the two observables associated with this technique, one is assessed to be highly perceivable (Network Traffic Associated with Work (Sending of Password List, Querying of Users, Communication Between Targets)).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                            |
|---------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 16 artifacts could be generated by the Valid Accounts technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                               |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                             |

### 3.18. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR LATERAL MOVEMENT

The Exploitation of Remote Services technique (T0866) is seen in Conficker's use of [MS08-067](#), otherwise known as [CVE-2008-4250](#), to attack other uninfected machines.<sup>38</sup> This exploit makes use of a RCE Vulnerability present in the RPC component of Microsoft Windows Server Service.<sup>39</sup> This technique is only effective against computers that have not applied the patch for the Microsoft Windows Server Service RPC Vulnerability ([BID 31874](#)). By targeting TCP Port 445, this exploit allows an adversary to trigger an overflow in the path canonicalization functionality of RPC and execute arbitrary code on a remote machine via use of a crafted RPC request.<sup>40</sup> If the malware successfully exploits the issue, the worm then creates an HTTP server on the compromised computer on a random port.<sup>41</sup>

If a network baseline is properly setup and monitoring capabilities are in place, OT Cybersecurity, IT Cybersecurity, and IT Staff may observe Event Viewer Log Entries.

A total of five observables were identified with the use of the [Exploitation of Remote Services technique \(T0866\)](#). This technique is important for investigation as software vulnerabilities are a common way for adversaries to take advantage of a programming error to enable remote service abuse. This technique appears late in the timeline and responding to patches released with vulnerabilities will limit spread to other networks.

Of the five observables associated with this technique, three are assessed to be highly perceivable (RPC Traffic; Targeting TCP (445); Event Viewer Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                             |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 31 artifacts could be generated by the Exploitation of Remote Services technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                                |
| <b>Resources</b>                                              | <a href="#">Technique Detection References</a>                                              |

### 3.19. PROGRAM DOWNLOAD TECHNIQUE (T0843) FOR LATERAL MOVEMENT

Before the download of scareware software, Conficker reads 0x400 bytes, and if the buffer is null-terminated, the malware passes the message to the function `thread_download_file_from_url` using Port 80/TCP (HTTP).<sup>42</sup> The message is interpreted as a string representing a URL (<https://trafficconverter.biz/4vir/antispyware/>) used to download an executable, `loadadv.exe`. Conficker then downloads two malicious files within this executable, `W.32 Waledac` and `SpywareProtect2009`.<sup>43</sup> `Waledac` is used to send spam via email, download and execute arbitrary files, harvest email addresses from the infected machine, proxy network traffic, sniff passwords, and perform denial of service attacks.<sup>44</sup> `SpyProtect2009` is a form of scareware that provides users with false indication that their computers were infected with malware.<sup>45</sup> It then encourages users to pay for the product to remove these phantom threats from the machine.<sup>46</sup>

If proper configurations are in place, OT Cybersecurity, IT Cybersecurity, and IT Staff may observe Event Viewer Log Entries.

A total of four observables were identified with the use of the [Program Download technique \(T0843\)](#). This technique is important for investigation since it allows adversaries to transfer a malicious program to end user devices. This technique appears later in the timeline and responding to it will prevent malicious payloads from propagating to additional machines. This technique modifies the host operating system files via the installation of `Waledac` and `SpyProtect2009`, and modifies system registry entries, resulting in the host being placed into a modified or compromised state. If system backups are created after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the four observables associated with this technique, three are assessed to be highly perceivable (File download: `loadadv.exe`; Web Connection to <https://trafficconverter.biz/4vir/antispyware/>; Event Viewer Log Entries).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                              |
|---------------------------------------------------------------|------------------------------------------------------------------------------|
| Artifacts                                                     | A total of 19 artifacts could be generated by the Program Download technique |
| Technique Observers                                           | OT Cybersecurity, IT Cybersecurity, IT Staff                                 |
| Resources                                                     | <a href="#">Technique Detection References</a>                               |

### 3.20. MANIPULATION OF CONTROL TECHNIQUE (T0831) FOR IMPACT

Conficker uses the Manipulation of Control technique (T0831) after its infection phase is complete. The Conficker malware connects to an external domain to add the newly infected computer to a list among several linked botnets. Once Conficker establishes a foothold it creates a bi-directional pipe (CreateNamedPipe API) where both the server and clients can write and read streams of messages limited to 0x400 bytes. The attacker could execute arbitrary code at this point or make changes to set point values, tags, or other parameters. The malware can also manipulate and disable Windows services related to updates, security, and removal products by denying users web access. This technique can be observed by logging and log analysis at both the network and host levels. Firewall, router, and host logs (including server logs) could all produce indicators of attack.

IT staff, OT Cybersecurity, and IT Cybersecurity personnel could observe this technique by establishing a network baseline and monitoring network traffic.

Two observables were identified with the [Manipulation of Control technique \(T0831\)](#). This technique is important for investigation as it may present an impact for the end-users or consumers of products and services. Terminating the chain of techniques at this point would likely limit the ability of the malware to spread and cause negative impacts to other systems, as well as limit the overall utility of the malware campaign by limiting individual nodes from executing unauthorized commands.

The two observables associated with this technique are assessed to be highly perceivable (Execution of Arbitrary Code through Botnet Configuration; CreateNamedPipe API).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                     |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of 16 artifacts could be generated by the Manipulation of Control technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff                                        |
| <b>Resources</b>                                              | <a href="#">Technique Detection Capability Sheet</a>                                |

### 3.21. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT

The Loss of Productivity and Revenue technique (T0828) represents the computer system shutdown required to contain and sanitize the malware. Gundremmingen was taken offline for two days after the discovery of the Conficker infection to perform scanning and mitigating security operations. This shutdown resulted in a loss of \$2.6 million in revenue.<sup>47</sup>

Because of the magnitude and global proliferation of Conficker impacts, the potential worldwide cost of finding and mitigating the malware is estimated at \$9.1 billion or more.<sup>48</sup> If the malware is not mitigated, Conficker could be leveraged by additional adversaries with potentially catastrophic results. For example, in 2011, a hacking gang from Ukraine used Conficker to trick people into buying fake anti-virus products, costing victims a total of \$72 million dollars.<sup>49</sup>

OT Cybersecurity, IT Cybersecurity, IT Staff, Engineering, OT Staff, Management, and Support Staff would perceive this technique through its impact on productivity and revenue loss.

Three observables were identified with the use of the [Loss of Productivity and Revenue technique \(T0828\)](#). This technique is important for investigation as it may present an impact for the end-users or consumers of products and services.

The three observables associated with this technique are assessed to be highly perceivable (Loss of Two Days of Productivity; Cost of “Spyware” Antivirus; Estimated Global Cost of \$9.1 Billion).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of five artifacts could be generated by the Loss of Productivity and Revenue technique |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff, Engineering, OT Staff, Management, Support Staff |
| <b>Resources</b>                                              | <a href="#">Technique Detection Capability Sheet and Recipe</a>                                |

### 3.22. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT

The Loss of Availability technique (T0826) occurred during a routine security audit that discovered the malware in the IT network of the Block B fuel IT computers.<sup>50</sup> Though it never reached the SCADA systems of the plant, officials decided to follow their predetermined security procedures and shut down the affected systems for two days. They used antivirus software to scan the plant's other systems and rolled systems to their last safe backup. All staff would have been able to observe the emergency shutdown system initiation.

OT Cybersecurity, IT Cybersecurity, IT Staff, Engineering, OT Staff, Management, and Support Staff would perceive this technique through its impact on system access and availability.

Three observables were identified with the use of the [Loss of Availability technique \(T0826\)](#). This technique is important for investigation as it prevents owners and operators from delivering products or services. This technique appears after the triggering event.

All three observables associated with this technique are assessed to be highly perceivable (Anti-virus Alerts; Two-day Downtime Due to Reconfiguration; Rollback to Last Safe Backup).

Please see [Appendix A](#) for the list of observables.

| CyOTE Capabilities for Technique Perception and Comprehension |                                                                                                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Artifacts</b>                                              | A total of eight artifacts could be generated by the Loss of Availability technique            |
| <b>Technique Observers</b>                                    | OT Cybersecurity, IT Cybersecurity, IT Staff, Engineering, OT Staff, Management, Support Staff |
| <b>Resources</b>                                              | <a href="#">Technique Detection Capability Sheet and Recipe</a>                                |

## APPENDIX A: OBSERVABLES LIBRARY

The observables listed here were associated or believed to be associated with the use of Conficker at Gundremmingen Nuclear Power Plant.

| Observables Associated with Replication Through Removable Media Technique (T0847) |                               |
|-----------------------------------------------------------------------------------|-------------------------------|
| Observable 1                                                                      | Autorun.inf Execution Feature |
| Observable 2                                                                      | Drive Creation                |
| Observable 3                                                                      | File Creation                 |
| Observable 4                                                                      | Process Access                |
| Observable 5                                                                      | Autoplay Alert                |
| Observable 6                                                                      | Windows Event Log Entries     |

| Observables Associated with Exploitation of Remote Services Technique (T0866) |                                                                                               |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Observable 1                                                                  | Remote Procedure Call (RPC) Traffic                                                           |
| Observable 2                                                                  | Microsoft Windows Server Service RPC Handling Remote Code Execution Vulnerability (BID 31874) |
| Observable 3                                                                  | TCP Port 445 SMB Network File                                                                 |
| Observable 4                                                                  | Creation of HTTP Server on a Random Port                                                      |
| Observable 5                                                                  | Windows Event Log Entries                                                                     |

| Observables Associated with Remote Services Technique (T0886) |                                      |
|---------------------------------------------------------------|--------------------------------------|
| Observable 1                                                  | Use of Admin Network Share (ADMIN\$) |
| Observable 2                                                  | NetServerEnum Request                |
| Observable 3                                                  | Windows Event ID 4624: Login         |
| Observable 4                                                  | Event ID 4625: Login Failures        |

| Observables Associated with Native API Technique (T0834) |                                                        |
|----------------------------------------------------------|--------------------------------------------------------|
| Observable 1                                             | Netapi32.dll and Zeek dcerpc.log                       |
| Observable 2                                             | Remote Procedure Call (RPC) Logs in Windows Event Logs |
| Observable 3                                             | Visible on Network if SMB is not Encrypted             |
| Observable 4                                             | Connect Back HTTP Connection                           |
| Observable 5                                             | DNS Request Associated with Bad Domain                 |
| Observable 6                                             | New Domain into the Environment                        |

| Observables Associated with Scripting Technique (T0853) |                                          |
|---------------------------------------------------------|------------------------------------------|
| Observable 1                                            | Portable Executable (PE) Dropped to Disk |

**Observables Associated with Service Stop Technique (T0881)**

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| <b>Observable 1</b> | Denial of DNS Requests to Domains Containing Conficker Antivirus |
| <b>Observable 2</b> | DNS Request Timeout                                              |
| <b>Observable 3</b> | Increase in DNS Retransmissions                                  |
| <b>Observable 4</b> | Disables Automatic Backup Settings                               |
| <b>Observable 5</b> | Deletes Backup/Restore Points                                    |

**Observables Associated with Rootkit Technique (T0851)**

|                     |                                                         |
|---------------------|---------------------------------------------------------|
| <b>Observable 1</b> | Network Request Time out from Security Related Websites |
| <b>Observable 2</b> | Increase in DNS Retransmissions                         |
| <b>Observable 3</b> | Blocking of Predetermined DNS Requests                  |

**Observables Associated with Block Reporting Message Technique (T0804)**

|                     |                                              |
|---------------------|----------------------------------------------|
| <b>Observable 1</b> | Disable Windows Security Alert Notifications |
| <b>Observable 2</b> | Prevents Access to Safe Mode                 |
| <b>Observable 3</b> | Event Viewer Log Entries                     |

**Observables Associated with Alarm Suppression Technique (T0878)**

|                     |                                |
|---------------------|--------------------------------|
| <b>Observable 1</b> | Disabling of Security Software |
| <b>Observable 2</b> | Prevent Access to Safe Mode    |
| <b>Observable 3</b> | Event Viewer Log Entries       |

**Observables Associated with Masquerading Technique (T0849)**

|                     |                                                                         |
|---------------------|-------------------------------------------------------------------------|
| <b>Observable 1</b> | DLL Creation, Renaming, and Deletion                                    |
| <b>Observable 2</b> | Registered DLLs with DisplayName Set to Empty and Type Set to Invisible |

**Observables Associated with Modify Program Technique (T0889)**

|                     |                                                |
|---------------------|------------------------------------------------|
| <b>Observable 1</b> | External IP Address Connection                 |
| <b>Observable 2</b> | HTTP Server Creation                           |
| <b>Observable 3</b> | File Download to Update Conficker Malware Form |
| <b>Observable 4</b> | Event Viewer Log Entries                       |

**Observables Associated with Standard Application Layer Protocol Technique (T0869)**

|                     |                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Observable 1</b> | Microsoft Windows Server Service Remote Procedure Call (RPC) Handling Remote Code Execution Vulnerability (BID 31874) |
| <b>Observable 2</b> | Targeting TCP Port 445 (SMB)                                                                                          |
| <b>Observable 3</b> | External Network Connections                                                                                          |
| <b>Observable 4</b> | Connection to Malicious URL Domains for C2C                                                                           |
| <b>Observable 5</b> | Windows Event Log Entries                                                                                             |

**Observables Associated with Commonly Used Port Technique (T0885)**

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| <b>Observable 1</b> | Remote Procedure Call (RPC) Traffic                                                           |
| <b>Observable 2</b> | Microsoft Windows Server Service RPC Handling Remote Code Execution Vulnerability (BID 31874) |
| <b>Observable 3</b> | Targeting TCP Port 445 (SMB)                                                                  |
| <b>Observable 4</b> | Creation of HTTP Server                                                                       |
| <b>Observable 5</b> | Windows Event Log Entries                                                                     |

**Observables Associated with Remote System Discovery Technique (T0846)**

|                     |                   |
|---------------------|-------------------|
| <b>Observable 1</b> | High UDP Traffic  |
| <b>Observable 2</b> | Port 53/UDP (DNS) |

**Observables Associated with Remote System Information Discovery Technique (T0888)**

|                     |                      |
|---------------------|----------------------|
| <b>Observable 1</b> | HTTP Request         |
| <b>Observable 2</b> | Binary Updates Check |
| <b>Observable 3</b> | File Signature Check |

**Observables Associated with Remote Services Technique (T0886)**

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| <b>Observable 1</b> | Execution of rundll32.exe                                         |
| <b>Observable 2</b> | Identification of Admin Share or Interprocess Communication (IPC) |

**Observables Associated with Valid Accounts Technique (T0859)**

|                     |                               |
|---------------------|-------------------------------|
| <b>Observable 1</b> | Brute-Force Password Attempts |
| <b>Observable 2</b> | Associated Network Traffic    |

**Observables Associated with Exploitation of Remote Services Technique (T0866)**

|                     |                                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------|
| <b>Observable 1</b> | Remote Procedure Call (RPC) Traffic                                                           |
| <b>Observable 2</b> | Microsoft Windows Server Service RPC Handling Remote Code Execution Vulnerability (BID 31874) |
| <b>Observable 3</b> | Targeting TCP Port 445                                                                        |
| <b>Observable 4</b> | Creation of HTTP Server                                                                       |
| <b>Observable 5</b> | Event Viewer Log Entries                                                                      |

**Observables Associated with Program Download Technique (T0843)**

|                     |                                                                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Observable 1</b> | Port 80/TCP (HTTP)                                                                                                              |
| <b>Observable 2</b> | Web Connection to <a href="https://trafficconverter.biz/4vir/antispysware/">https://trafficconverter.biz/4vir/antispysware/</a> |
| <b>Observable 3</b> | File Download loadadv.exe                                                                                                       |
| <b>Observable 4</b> | Event Viewer Log Entries                                                                                                        |

**Observables Associated with Manipulation of Control Technique (T0831)**

|                     |                                                          |
|---------------------|----------------------------------------------------------|
| <b>Observable 1</b> | Execution of Arbitrary Code through Botnet Configuration |
| <b>Observable 2</b> | CreateNamedPipe API                                      |

**Observables Associated with Loss of Productivity and Revenue Technique (T0828)**

|                     |                                        |
|---------------------|----------------------------------------|
| <b>Observable 1</b> | Loss of Two Days Productivity          |
| <b>Observable 2</b> | Estimated Global Cost of \$9.1 Billion |
| <b>Observable 3</b> | Cost of "Spyware" Antivirus            |

**Observables Associated with Loss of Availability Technique (T0826)**

|                     |                                         |
|---------------------|-----------------------------------------|
| <b>Observable 1</b> | Two-day Downtime Due to Reconfiguration |
| <b>Observable 2</b> | Antivirus Alerts                        |
| <b>Observable 3</b> | Rollback to Last Safe Backup            |

## APPENDIX B: ARTIFACTS LIBRARY

| Artifacts Associated with Replication Through Removable Media Technique (T0847) |                                                                      |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Artifact 1</b>                                                               | ARP Connections                                                      |
| <b>Artifact 2</b>                                                               | FTP Port 21                                                          |
| <b>Artifact 3</b>                                                               | FTPS Port 990                                                        |
| <b>Artifact 4</b>                                                               | HTTPS Port 443                                                       |
| <b>Artifact 5</b>                                                               | HTTP Port 80                                                         |
| <b>Artifact 6</b>                                                               | USB Make                                                             |
| <b>Artifact 7</b>                                                               | USB Model                                                            |
| <b>Artifact 8</b>                                                               | USB Version                                                          |
| <b>Artifact 9</b>                                                               | USB Serial Number                                                    |
| <b>Artifact 10</b>                                                              | Mass Storage Class (MSC) Connections                                 |
| <b>Artifact 11</b>                                                              | Picture Transfer Protocol (PTP) Connections                          |
| <b>Artifact 12</b>                                                              | MAC Address                                                          |
| <b>Artifact 13</b>                                                              | Media Transfer Protocol (MTP) Connections                            |
| <b>Artifact 14</b>                                                              | .LNK File                                                            |
| <b>Artifact 15</b>                                                              | Network Discovery Protocols Traffic                                  |
| <b>Artifact 16</b>                                                              | Changes to System Registry SYSTEM\CurrentControlSet\Enum\USBSTOR     |
| <b>Artifact 17</b>                                                              | Drive Letter Creation                                                |
| <b>Artifact 18</b>                                                              | Changes to System Registry SYSTEM\MountedDevices                     |
| <b>Artifact 19</b>                                                              | Bytes sent from System Resource Usage Manager (SRUM)                 |
| <b>Artifact 20</b>                                                              | Bytes Received from SRUM                                             |
| <b>Artifact 21</b>                                                              | Device User (USB)                                                    |
| <b>Artifact 22</b>                                                              | Last Time Device Connected (USB)                                     |
| <b>Artifact 23</b>                                                              | Source IP Address                                                    |
| <b>Artifact 24</b>                                                              | First Time Device Connected (USB)                                    |
| <b>Artifact 25</b>                                                              | Device Disconnected Time (USB)                                       |
| <b>Artifact 26</b>                                                              | Wireless Transmission                                                |
| <b>Artifact 27</b>                                                              | Plug and Play Log File setupapi.log                                  |
| <b>Artifact 28</b>                                                              | System Log Plug and Play Driver Installed Event 20001                |
| <b>Artifact 29</b>                                                              | Security Log Attempt to Access Removable Storage Object Event (4663) |
| <b>Artifact 30</b>                                                              | Security Log Failure to Access Removeable Device (4656)              |
| <b>Artifact 31</b>                                                              | Network Connections with Honeypot                                    |
| <b>Artifact 32</b>                                                              | Honey Pot Logs                                                       |
| <b>Artifact 33</b>                                                              | User Agents                                                          |

| Artifacts Associated with Replication Through Removable Media Technique (T0847) |                                    |
|---------------------------------------------------------------------------------|------------------------------------|
| <b>Artifact 34</b>                                                              | DNS Queries Traffic                |
| <b>Artifact 35</b>                                                              | File Access Created                |
| <b>Artifact 36</b>                                                              | File Access Removed                |
| <b>Artifact 37</b>                                                              | Process Creation                   |
| <b>Artifact 38</b>                                                              | Process Termination                |
| <b>Artifact 39</b>                                                              | Device Failure                     |
| <b>Artifact 40</b>                                                              | Device Reboot                      |
| <b>Artifact 41</b>                                                              | Performance Degradation            |
| <b>Artifact 42</b>                                                              | Command Line Dialog Box Opened     |
| <b>Artifact 43</b>                                                              | Recent Connections Metadata Change |
| <b>Artifact 44</b>                                                              | RDP Traffic Port 3389              |
| <b>Artifact 45</b>                                                              | VNC Traffic Port 5900              |
| <b>Artifact 46</b>                                                              | SSH Traffic Port 22                |
| <b>Artifact 47</b>                                                              | Telnet Traffic                     |
| <b>Artifact 48</b>                                                              | TFTP Port 69                       |

| Artifacts Associated with Exploitation of Remote Services Technique (T0866) |                                         |
|-----------------------------------------------------------------------------|-----------------------------------------|
| <b>Artifact 1</b>                                                           | Application Logs                        |
| <b>Artifact 2</b>                                                           | Connection to HMI End Points            |
| <b>Artifact 3</b>                                                           | Connection to EWS End Points            |
| <b>Artifact 4</b>                                                           | Connection to Data Historian End Points |
| <b>Artifact 5</b>                                                           | Connection to Controller End Points     |
| <b>Artifact 6</b>                                                           | Manipulation of Process                 |
| <b>Artifact 7</b>                                                           | Manipulation of Set Points              |
| <b>Artifact 8</b>                                                           | Misconfigurations of End Points         |
| <b>Artifact 9</b>                                                           | Process Failure                         |
| <b>Artifact 10</b>                                                          | Controller Failure                      |
| <b>Artifact 11</b>                                                          | Code Injections into Application        |
| <b>Artifact 12</b>                                                          | Application Log on Event                |
| <b>Artifact 13</b>                                                          | Code Injection into the OS              |
| <b>Artifact 14</b>                                                          | OPC Code Injection                      |
| <b>Artifact 15</b>                                                          | Database Command Executions             |
| <b>Artifact 16</b>                                                          | User Events Across Multiple Devices     |
| <b>Artifact 17</b>                                                          | Host System Registry Changes            |

| Artifacts Associated with Exploitation of Remote Services Technique (T0866) |                                         |
|-----------------------------------------------------------------------------|-----------------------------------------|
| <b>Artifact 18</b>                                                          | Security Events Across Multiple Devices |
| <b>Artifact 19</b>                                                          | Kernel Level Events                     |
| <b>Artifact 20</b>                                                          | System Reboots                          |
| <b>Artifact 21</b>                                                          | Blank Screens                           |
| <b>Artifact 22</b>                                                          | Safemode Reboot                         |
| <b>Artifact 23</b>                                                          | Application Log off Event               |
| <b>Artifact 24</b>                                                          | Alarm Events                            |
| <b>Artifact 25</b>                                                          | Absence of Alarm Events                 |
| <b>Artifact 26</b>                                                          | Common Network Traffic                  |
| <b>Artifact 27</b>                                                          | Remote Network Traffic                  |
| <b>Artifact 28</b>                                                          | Vendor Specific Network Traffic         |
| <b>Artifact 29</b>                                                          | Industrial Protocol Network Traffic     |
| <b>Artifact 30</b>                                                          | SQL Protocol                            |
| <b>Artifact 31</b>                                                          | SMB Protocol                            |

| Artifacts Associated with Remote Services Technique (T0886) |                                   |
|-------------------------------------------------------------|-----------------------------------|
| <b>Artifact 1</b>                                           | Remote Client Connection          |
| <b>Artifact 2</b>                                           | Logon Event                       |
| <b>Artifact 3</b>                                           | Logoff                            |
| <b>Artifact 4</b>                                           | Logoff Event                      |
| <b>Artifact 5</b>                                           | Registry Changes                  |
| <b>Artifact 6</b>                                           | Registry Connection Change        |
| <b>Artifact 7</b>                                           | Mouse Movement                    |
| <b>Artifact 8</b>                                           | Unexpected I/O                    |
| <b>Artifact 9</b>                                           | Desktop Prompt Windows Created    |
| <b>Artifact 10</b>                                          | Session Cache                     |
| <b>Artifact 11</b>                                          | Application Log                   |
| <b>Artifact 12</b>                                          | RDP Traffic Port 3389             |
| <b>Artifact 13</b>                                          | System Log Event                  |
| <b>Artifact 14</b>                                          | Authentication Logs               |
| <b>Artifact 15</b>                                          | GUI Modifications                 |
| <b>Artifact 16</b>                                          | Data File Size in Network Content |
| <b>Artifact 17</b>                                          | File Movement                     |
| <b>Artifact 18</b>                                          | MSSQL Traffic Port 1433           |

| Artifacts Associated with Remote Services Technique (T0886) |                                   |
|-------------------------------------------------------------|-----------------------------------|
| <b>Artifact 19</b>                                          | SSH Traffic Port 22               |
| <b>Artifact 20</b>                                          | SMB Traffic Ports 139, 445        |
| <b>Artifact 21</b>                                          | VNC Traffic Ports 5800, 5900      |
| <b>Artifact 22</b>                                          | Process Creation                  |
| <b>Artifact 23</b>                                          | Remote Session Creation Timestamp |
| <b>Artifact 24</b>                                          | Network Traffic Content Creation  |

| Artifacts Associated with Native API Technique (T0834) |                                          |
|--------------------------------------------------------|------------------------------------------|
| <b>Artifact 1</b>                                      | Industrial Network Traffic               |
| <b>Artifact 2</b>                                      | Industrial Protocol Command Packet       |
| <b>Artifact 3</b>                                      | Device Reads                             |
| <b>Artifact 4</b>                                      | Device I/O Image Table Manipulated       |
| <b>Artifact 5</b>                                      | Device Failure                           |
| <b>Artifact 6</b>                                      | Alter Process Logic                      |
| <b>Artifact 7</b>                                      | Device Performance Degradation           |
| <b>Artifact 8</b>                                      | Device Memory Modification               |
| <b>Artifact 9</b>                                      | Device Alarm                             |
| <b>Artifact 10</b>                                     | Device Live Data Changes                 |
| <b>Artifact 11</b>                                     | Systems Calls                            |
| <b>Artifact 12</b>                                     | Alert Generated                          |
| <b>Artifact 13</b>                                     | Memory Corruption                        |
| <b>Artifact 14</b>                                     | Host Device Failure                      |
| <b>Artifact 15</b>                                     | Blue Screen                              |
| <b>Artifact 16</b>                                     | Performance Degradation                  |
| <b>Artifact 17</b>                                     | Sysmon Events Created                    |
| <b>Artifact 18</b>                                     | Services Initiated                       |
| <b>Artifact 19</b>                                     | Processes Initiated                      |
| <b>Artifact 20</b>                                     | Files Created                            |
| <b>Artifact 21</b>                                     | Imports Hash Changed                     |
| <b>Artifact 22</b>                                     | .DLL Modifications                       |
| <b>Artifact 23</b>                                     | System Resource Usage Management Changes |
| <b>Artifact 24</b>                                     | Command Execution                        |
| <b>Artifact 25</b>                                     | Configuration Change                     |

| Artifacts Associated with Scripting Technique (T0853) |                              |
|-------------------------------------------------------|------------------------------|
| <b>Artifact 1</b>                                     | Files Dropped into Directory |
| <b>Artifact 2</b>                                     | System Event Log Creation    |
| <b>Artifact 3</b>                                     | OS Timeline Event            |
| <b>Artifact 4</b>                                     | Startup Menu Modification    |
| <b>Artifact 5</b>                                     | System Processes Created     |
| <b>Artifact 6</b>                                     | Windows API Event Log        |
| <b>Artifact 7</b>                                     | Executable Files             |
| <b>Artifact 8</b>                                     | Prefetch Files Created       |
| <b>Artifact 9</b>                                     | External Network Connections |
| <b>Artifact 10</b>                                    | Network Services Created     |
| <b>Artifact 11</b>                                    | Registry Modifications       |
| <b>Artifact 12</b>                                    | OS Service Installation      |

| Artifacts Associated with Service Stop Technique (T0881) |                                                        |
|----------------------------------------------------------|--------------------------------------------------------|
| <b>Artifact 1</b>                                        | Process Failure                                        |
| <b>Artifact 2</b>                                        | Alarm Event                                            |
| <b>Artifact 3</b>                                        | Internal System Logs                                   |
| <b>Artifact 4</b>                                        | Application Error Messages                             |
| <b>Artifact 5</b>                                        | Process Error Messages                                 |
| <b>Artifact 6</b>                                        | Application Service Stop                               |
| <b>Artifact 7</b>                                        | OS Service Crash                                       |
| <b>Artifact 8</b>                                        | System Event Logs                                      |
| <b>Artifact 9</b>                                        | Application Event Logs                                 |
| <b>Artifact 10</b>                                       | OS API Call                                            |
| <b>Artifact 11</b>                                       | Command Line System Argument                           |
| <b>Artifact 12</b>                                       | System Resource Usage Manager Application Usage Change |
| <b>Artifact 13</b>                                       | Registry Change HKLM\SYSTEM\currentcontrolset\services |

| Artifacts Associated with Rootkit Technique (T0851) |                                                      |
|-----------------------------------------------------|------------------------------------------------------|
| <b>Artifact 1</b>                                   | Drive Modification                                   |
| <b>Artifact 2</b>                                   | Industrial Network Traffic for Remote Code Execution |
| <b>Artifact 3</b>                                   | Remote Network Traffic for Remote Code Execution     |
| <b>Artifact 4</b>                                   | Modified Reporting                                   |
| <b>Artifact 5</b>                                   | Failed Reporting                                     |

| Artifacts Associated with Rootkit Technique (T0851) |                                                       |
|-----------------------------------------------------|-------------------------------------------------------|
| <b>Artifact 6</b>                                   | Blue Screen on Workstation                            |
| <b>Artifact 7</b>                                   | System Resource Usage Monitor (SRUM) Metadata Changes |
| <b>Artifact 8</b>                                   | Performance Mismatched with Performance Reporting     |
| <b>Artifact 9</b>                                   | Packet Mismatch Between Network and Host Sources      |
| <b>Artifact 10</b>                                  | Unusual Protocol Field Values                         |
| <b>Artifact 11</b>                                  | Increased Maintenance Without Faults Reported         |
| <b>Artifact 12</b>                                  | Firmware Number Modification                          |
| <b>Artifact 13</b>                                  | Blocked Read Requests at OS Level                     |
| <b>Artifact 14</b>                                  | Blocked Delete Requests at OS Level                   |
| <b>Artifact 15</b>                                  | Blocked Create Requests at OS Level                   |
| <b>Artifact 16</b>                                  | Blocked Write Requests at OS Level                    |
| <b>Artifact 17</b>                                  | Firmware Flash Update                                 |
| <b>Artifact 18</b>                                  | Device Reboot                                         |
| <b>Artifact 19</b>                                  | Device Failure                                        |
| <b>Artifact 20</b>                                  | Dropped API Calls                                     |
| <b>Artifact 21</b>                                  | Application Logs                                      |
| <b>Artifact 22</b>                                  | Dialog Box Requesting Update                          |
| <b>Artifact 23</b>                                  | Common Network Traffic for Remote Code Execution      |

| Artifacts Associated with Block Reporting Message Technique (T0804) |                                                     |
|---------------------------------------------------------------------|-----------------------------------------------------|
| <b>Artifact 1</b>                                                   | Application Modification                            |
| <b>Artifact 2</b>                                                   | Physical Process Changes without Data Received      |
| <b>Artifact 3</b>                                                   | Conflicting Device Status Reports                   |
| <b>Artifact 4</b>                                                   | I/O Values Mismatched with Process Current State    |
| <b>Artifact 5</b>                                                   | Delayed Operational Process Status Change           |
| <b>Artifact 6</b>                                                   | Application Log Event Absent                        |
| <b>Artifact 7</b>                                                   | Historian Database Missing Data                     |
| <b>Artifact 8</b>                                                   | I/O Server Nonresponsive                            |
| <b>Artifact 9</b>                                                   | Real-Time Operational Data Missing                  |
| <b>Artifact 10</b>                                                  | Supervisory Application Logs Mismatch Current State |
| <b>Artifact 11</b>                                                  | Process Status Modification                         |
| <b>Artifact 12</b>                                                  | Network Traffic Changes                             |
| <b>Artifact 13</b>                                                  | Network Connections Creation                        |
| <b>Artifact 14</b>                                                  | Operational Device Failure                          |

**Artifacts Associated with Block Reporting Message Technique (T0804)**

|                    |                                           |
|--------------------|-------------------------------------------|
| <b>Artifact 15</b> | Operational Database Data Modification    |
| <b>Artifact 16</b> | Operational Database Configuration Change |
| <b>Artifact 17</b> | Operational Process Termination           |
| <b>Artifact 18</b> | Operational Process Alarm Failures        |

**Artifacts Associated with Alarm Suppression Technique (T0878)**

|                    |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| <b>Artifact 1</b>  | Modification of Alarm Set Points                                           |
| <b>Artifact 2</b>  | Runaway Process State                                                      |
| <b>Artifact 3</b>  | Catastrophic Failures                                                      |
| <b>Artifact 4</b>  | Configuration Change Logs                                                  |
| <b>Artifact 5</b>  | Increased Number of Output Quality Assurance Failures                      |
| <b>Artifact 6</b>  | Insertion of Malicious Industrial Protocol to Suppress True Process Values |
| <b>Artifact 7</b>  | Modification of SQL Database Inputs                                        |
| <b>Artifact 8</b>  | SQL Protocol Network Traffic                                               |
| <b>Artifact 9</b>  | External Connection to Operational Database                                |
| <b>Artifact 10</b> | Mismatch Between Sensor Reporting and Physical Process                     |
| <b>Artifact 11</b> | Change in Process Output                                                   |
| <b>Artifact 12</b> | Increased Maintenance Issues                                               |
| <b>Artifact 13</b> | Control System Degradation                                                 |

**Artifacts Associated with Masquerading Technique (T0849)**

|                    |                                                    |
|--------------------|----------------------------------------------------|
| <b>Artifact 1</b>  | File Creation with Common Name                     |
| <b>Artifact 2</b>  | Additional File Directories Created                |
| <b>Artifact 3</b>  | Scheduled Job Modification                         |
| <b>Artifact 4</b>  | Service Creation                                   |
| <b>Artifact 5</b>  | Services Metadata                                  |
| <b>Artifact 6</b>  | Scheduled Job Metadata                             |
| <b>Artifact 7</b>  | Leetspeak User Metadata                            |
| <b>Artifact 8</b>  | Common Application with Non-Native Child Processes |
| <b>Artifact 9</b>  | Process Metadata Changes                           |
| <b>Artifact 10</b> | Command Line Execution                             |
| <b>Artifact 11</b> | File Modification                                  |
| <b>Artifact 12</b> | Warez Application Use                              |
| <b>Artifact 13</b> | Leetspeak File Creation                            |

**Artifacts Associated with Masquerading Technique (T0849)**

|                    |                                          |
|--------------------|------------------------------------------|
| <b>Artifact 14</b> | Applications Causing Unintended Actions  |
| <b>Artifact 15</b> | Additional Functionality in Applications |

**Artifacts Associated with Modify Program Technique (T0889)**

|                   |                                                                     |
|-------------------|---------------------------------------------------------------------|
| <b>Artifact 1</b> | Unexpected Program Download Observed on Network                     |
| <b>Artifact 2</b> | Modification to Application Responsible for Program Downloads       |
| <b>Artifact 3</b> | Unexpected Modification to Program Organizational Units on a Device |

**Artifacts Associated with Standard Application Layer Protocol Technique (T0869)**

|                    |                                                |
|--------------------|------------------------------------------------|
| <b>Artifact 1</b>  | External Network Connections                   |
| <b>Artifact 2</b>  | DNS Autonomous System Number                   |
| <b>Artifact 3</b>  | Increase in the Number of External Connections |
| <b>Artifact 4</b>  | Network Content Metadata                       |
| <b>Artifact 5</b>  | Network Connection Times                       |
| <b>Artifact 6</b>  | HTTP Traffic Port 80                           |
| <b>Artifact 7</b>  | DNS Traffic Port 53                            |
| <b>Artifact 8</b>  | SMB Traffic Port 445                           |
| <b>Artifact 9</b>  | HTTPS Traffic Port 443                         |
| <b>Artifact 10</b> | RDP Traffic Port 3389                          |
| <b>Artifact 11</b> | HTTP Post Request                              |
| <b>Artifact 12</b> | External IP Addresses                          |

**Artifacts Associated with Commonly Used Port Technique (T0885)**

|                   |                                                                            |
|-------------------|----------------------------------------------------------------------------|
| <b>Artifact 1</b> | Unexpected Process Usage of Common Port Observed via OS Commands (Netstat) |
| <b>Artifact 2</b> | Unexpected Process Usage of Common Port Observed via Memory                |
| <b>Artifact 3</b> | Unexpected Process Usage of Common Port Observed via OS Logs               |
| <b>Artifact 4</b> | Unexpected Process Usage of Common Port Observed via Firewall Logs         |
| <b>Artifact 5</b> | Unexpected Host Communicating with Common Port on Industrial Asset         |

**Artifacts Associated with Remote System Discovery Technique (T0846)**

|                   |                                     |
|-------------------|-------------------------------------|
| <b>Artifact 1</b> | Common Network Traffic              |
| <b>Artifact 2</b> | IEC 103 Traffic (for North America) |
| <b>Artifact 3</b> | IEC 61850 MMS and GOOSE             |

| Artifacts Associated with Remote System Discovery Technique (T0846) |                                                                   |
|---------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Artifact 4</b>                                                   | Controller Proprietary Traffic                                    |
| <b>Artifact 5</b>                                                   | Echo Type 8 Traffic                                               |
| <b>Artifact 6</b>                                                   | ICMP Type 7 Traffic                                               |
| <b>Artifact 7</b>                                                   | SNMP Port 162 Traffic                                             |
| <b>Artifact 8</b>                                                   | SNMP Port 161 Traffic                                             |
| <b>Artifact 9</b>                                                   | Command Line Dialog Box Open                                      |
| <b>Artifact 10</b>                                                  | Operating System Queries                                          |
| <b>Artifact 11</b>                                                  | DNS Port 53 Zone Transfers                                        |
| <b>Artifact 12</b>                                                  | Industrial Network Traffic Content about Hostnames                |
| <b>Artifact 13</b>                                                  | Polling Network Traffic from Unauthorized IP Sender Addresses     |
| <b>Artifact 14</b>                                                  | NetBIOS Name Services Port 137                                    |
| <b>Artifact 15</b>                                                  | LDAP Port 389                                                     |
| <b>Artifact 16</b>                                                  | Active Directory Calls                                            |
| <b>Artifact 17</b>                                                  | Email Server Calls                                                |
| <b>Artifact 18</b>                                                  | SMTP Port 25 Traffic                                              |
| <b>Artifact 19</b>                                                  | DNS Lookup Queries                                                |
| <b>Artifact 20</b>                                                  | ARP Scans                                                         |
| <b>Artifact 21</b>                                                  | TCP Connect Scan                                                  |
| <b>Artifact 22</b>                                                  | TCP SYN Scans                                                     |
| <b>Artifact 23</b>                                                  | Scans over Industrial Network Ports with Target IPs               |
| <b>Artifact 24</b>                                                  | TCP FIN Scans                                                     |
| <b>Artifact 25</b>                                                  | TCP Reverse Ident Scan                                            |
| <b>Artifact 26</b>                                                  | TCP XMAS Scan                                                     |
| <b>Artifact 27</b>                                                  | TCP ACK Scan                                                      |
| <b>Artifact 28</b>                                                  | VNC Port 5900 Calls                                               |
| <b>Artifact 29</b>                                                  | Protocol content Enumeration                                      |
| <b>Artifact 30</b>                                                  | Protocol Header Enumeration                                       |
| <b>Artifact 31</b>                                                  | Recurring Protocol SYN Traffic                                    |
| <b>Artifact 32</b>                                                  | Sequential Protocol SYN Traffic                                   |
| <b>Artifact 33</b>                                                  | Statistical anomalies in network traffic                          |
| <b>Artifact 34</b>                                                  | Industrial Network Traffic Content Containing Logical Identifiers |
| <b>Artifact 35</b>                                                  | Device Failure                                                    |
| <b>Artifact 36</b>                                                  | Device Reboot                                                     |
| <b>Artifact 37</b>                                                  | Bandwidth Degradation                                             |
| <b>Artifact 38</b>                                                  | Host Recent Connection Logs                                       |

#### Artifacts Associated with Remote System Discovery Technique (T0846)

|                    |                                   |
|--------------------|-----------------------------------|
| <b>Artifact 39</b> | Industrial Network Traffic        |
| <b>Artifact 40</b> | OPC Network Traffic               |
| <b>Artifact 41</b> | IEC 104                           |
| <b>Artifact 42</b> | IEC 102                           |
| <b>Artifact 43</b> | IEC 101 Traffic to Serial Devices |

#### Artifacts Associated with Remote System Information Discovery Technique (T0888)

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| <b>Artifact 1</b> | Unexpected Industrial Protocol Usage                                                  |
| <b>Artifact 2</b> | Unexpected Industrial Application Usage                                               |
| <b>Artifact 3</b> | Unexpected Standard Protocol Usage                                                    |
| <b>Artifact 4</b> | Unexpected Recon Associated Command Line Options (Ping Sweep, Netstat, etc.)          |
| <b>Artifact 5</b> | Unexpected Recon Associated Child Processes (Ping Sweep, Netstat, Etc.)               |
| <b>Artifact 6</b> | Unexpected Recon Associated Library Calls                                             |
| <b>Artifact 7</b> | Exfiltration of Host, Network, and/or System Architecture or Configuration Data       |
| <b>Artifact 8</b> | Compromise and Exfiltration of Data from Asset Information Datastores or Applications |

#### Artifacts Associated with Valid Accounts Technique (T0859)

|                    |                                        |
|--------------------|----------------------------------------|
| <b>Artifact 1</b>  | Logons                                 |
| <b>Artifact 2</b>  | Default Credential Use                 |
| <b>Artifact 3</b>  | Application Log                        |
| <b>Artifact 4</b>  | Domain Permission Requests             |
| <b>Artifact 5</b>  | Permission Elevation Requests          |
| <b>Artifact 6</b>  | Application Use Times                  |
| <b>Artifact 7</b>  | Configuration Changes                  |
| <b>Artifact 8</b>  | Prefetch Files Created After Execution |
| <b>Artifact 9</b>  | Logon Session Creation                 |
| <b>Artifact 10</b> | User Account Creation                  |
| <b>Artifact 11</b> | Authentication Creation                |
| <b>Artifact 12</b> | System Logs                            |
| <b>Artifact 13</b> | Successful Logon Event ID: 4624        |
| <b>Artifact 14</b> | Failed Logons Event ID: 4625           |
| <b>Artifact 15</b> | Logon Timestamp                        |
| <b>Artifact 16</b> | Logon Type Entry                       |

| Artifacts Associated with Program Download Technique (T0843) |                                                        |
|--------------------------------------------------------------|--------------------------------------------------------|
| <b>Artifact 1</b>                                            | Controller in Stop State                               |
| <b>Artifact 2</b>                                            | Operational Process Restart                            |
| <b>Artifact 3</b>                                            | Operational Database Data Modification                 |
| <b>Artifact 4</b>                                            | Device Alert                                           |
| <b>Artifact 5</b>                                            | Device Alarm                                           |
| <b>Artifact 6</b>                                            | Controller Application Log Event                       |
| <b>Artifact 7</b>                                            | Supervisory Workstation Program Download Popup         |
| <b>Artifact 8</b>                                            | Controller Application Log Type                        |
| <b>Artifact 9</b>                                            | Controller Application Log Timestamp                   |
| <b>Artifact 10</b>                                           | Controller Network Connections via Management Protocol |
| <b>Artifact 11</b>                                           | Controller Connection to External Website              |
| <b>Artifact 12</b>                                           | Controller in Program State                            |
| <b>Artifact 13</b>                                           | Controller Connected to External Networks              |
| <b>Artifact 14</b>                                           | Network Traffic Creation                               |
| <b>Artifact 15</b>                                           | Network Metadata                                       |
| <b>Artifact 16</b>                                           | External Domain Connection                             |
| <b>Artifact 17</b>                                           | External IP Address                                    |
| <b>Artifact 18</b>                                           | Controller State Change                                |
| <b>Artifact 19</b>                                           | Operational Process Shutdown                           |

| Artifacts Associated with Manipulation of Control Technique (T0831) |                                        |
|---------------------------------------------------------------------|----------------------------------------|
| <b>Artifact 1</b>                                                   | HMI Input Manipulation                 |
| <b>Artifact 2</b>                                                   | Command Execution                      |
| <b>Artifact 3</b>                                                   | Event Log Creation                     |
| <b>Artifact 4</b>                                                   | Application Log Event                  |
| <b>Artifact 5</b>                                                   | Altered Command Sequences              |
| <b>Artifact 6</b>                                                   | Application File Modification          |
| <b>Artifact 7</b>                                                   | Operational Data Modification          |
| <b>Artifact 8</b>                                                   | I/O Modification                       |
| <b>Artifact 9</b>                                                   | Engineering Workstation Mouse Movement |
| <b>Artifact 10</b>                                                  | Controller Set Point Change            |
| <b>Artifact 11</b>                                                  | Controller Parameter Change            |
| <b>Artifact 12</b>                                                  | Controller Tag Change                  |

**Artifacts Associated with Manipulation of Control Technique (T0831)**

|                    |                      |
|--------------------|----------------------|
| <b>Artifact 13</b> | Process State Change |
| <b>Artifact 14</b> | Process Shutdown     |
| <b>Artifact 15</b> | Process Restart      |
| <b>Artifact 16</b> | Process Initiated    |

**Artifacts Associated with Loss of Productivity and Revenue Technique (T0828)**

|                   |                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Artifact 1</b> | Loss of Confidence in a Safety System due to Unreliability Might Result in a Risk Management-Driven Shutdown of a Plant                      |
| <b>Artifact 2</b> | Wormable or Other Highly Propagating Malware Might Result in the Shutdown of a Plant to Prevent Ransomware or Other Destructive Attacks      |
| <b>Artifact 3</b> | Extortion Attempts Might Lead to Reduced Operations due to Potential Presence of Malicious Attackers                                         |
| <b>Artifact 4</b> | Loss of Control of Critical Systems Due to Ransomware or Loss of Confidence Might Lead to a Degraded Productivity or Revenue Operating State |
| <b>Artifact 5</b> | File System Modification Artifacts Might be Associated with the Loss of Productivity and Revenue Attack might be Present on Disk             |

**Artifacts Associated with Loss of Availability Technique (T0826)**

|                   |                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Artifact 1</b> | Operator or User Discovery of Encrypted or Inoperable Systems                                                 |
| <b>Artifact 2</b> | Significant Logged Usage of Native Crypto Functions or Presence of Import of Crypto Functions in Binaries     |
| <b>Artifact 3</b> | Significant Reduction or Increase in Network Traffic Due to Malware Propagation or Disappearance of Services  |
| <b>Artifact 4</b> | Unexplained Loss of Application Data                                                                          |
| <b>Artifact 5</b> | Unexplained Loss of User Data                                                                                 |
| <b>Artifact 6</b> | Process Failure Due to Loss of Required Network or System Dependency                                          |
| <b>Artifact 7</b> | Changes in Network Routing or Usage of Redundant Control System Network Connection Due to Failed Network Path |
| <b>Artifact 8</b> | File System Modification Artifacts Might Be Associated with the Loss of Availability Might be Present on Disk |

## APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in OT organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings](#) listed in the [SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering</b>  <ul style="list-style-type: none"><li>• Process Engineer</li><li>• Electrical, Controls, and Mechanical Engineer</li><li>• Project Engineer</li><li>• Systems and Reliability Engineer</li><li>• OT Developer</li><li>• PLC Programmer</li><li>• Emergency Operations Manager</li><li>• Plant Networking</li><li>• Control/Instrumentation Specialist</li><li>• Protection and Controls</li><li>• Field Engineer</li><li>• System Integrator</li></ul> | <b>Support Staff</b>  <ul style="list-style-type: none"><li>• Remote Maintenance &amp; Technical Support</li><li>• Contractors (engineering)</li><li>• IT and Physical Security Contractor</li><li>• Procurement Specialist</li><li>• Legal</li><li>• Contracting Engineer</li><li>• Insurance</li><li>• Supply-chain Participant</li><li>• Inventory Management/Lifecycle Management</li><li>• Physical Security Specialist</li></ul> |
| <b>Operations Technology (OT) Staff</b>  <ul style="list-style-type: none"><li>• Operator</li><li>• Site Security POC</li><li>• Technical Specialists (electrical/mechanical/chemical)</li><li>• ICS/SCADA Programmer</li></ul>                                                                                                                                                                                                                                            | <b>Information Technology (IT) Cybersecurity</b>  <ul style="list-style-type: none"><li>• ICS Security Analyst</li><li>• Security Engineering and Architect</li><li>• Security Operations</li><li>• Security Response and Forensics</li><li>• Security Management (CSO)</li><li>• Audit Specialist</li><li>• Security Tester</li></ul>                                                                                                |
| <b>Operational Technology (OT) Cybersecurity</b>  <ul style="list-style-type: none"><li>• OT Security</li><li>• ICS/SCADA Security</li></ul>                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Management</b>  <ul style="list-style-type: none"><li>• Plant Manager</li><li>• Risk/Safety Manager</li><li>• Business Unit Management</li><li>• C-level Management</li></ul>                                                                                                                                                                                                                                                                                         | <b>Information Technology (IT) Staff</b>  <ul style="list-style-type: none"><li>• Networking and Infrastructure</li><li>• Host Administrator</li><li>• Database Administrator</li><li>• Application Development</li><li>• ERP/MES Administrator</li><li>• IT Management</li></ul>                                                                                                                                                   |

## REFERENCES

- <sup>1</sup> [SENKI | The Rendon Group | “Conficker Working Group: Lessons Learned” | [https://www.senki.org/wp-content/uploads/2020/11/Conficker-Working-Group-Lessons-Learned-June-2010-Published-January-2011-whitepaper\\_76813745321.pdf](https://www.senki.org/wp-content/uploads/2020/11/Conficker-Working-Group-Lessons-Learned-June-2010-Published-January-2011-whitepaper_76813745321.pdf) | January 2011 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>2</sup> [Reuters | Christoph Steitz and Eric Auchard | “German nuclear plant infected with computer viruses, operator says” | <https://www.reuters.com/article/us-nuclearpower-cyber-germany-idUKKCN0XN2OS> | 26 April 2016 | Accessed 1 July 2022 | The source is publicly available information and does not contain classification markings]
- <sup>3</sup> [Softpedia | Catalin Cimpanu | “Malware Shuts Down German Nuclear Power Plant on Chernobyl’s 30<sup>th</sup> Anniversary” | <https://news.softpedia.com/news/on-chernobyl-s-30th-anniversary-malware-shuts-down-german-nuclear-power-plant-503429.shtml> | 26 April 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>4</sup> [Reuters | Christoph Steitz and Eric Auchard | “German nuclear plant infected with computer viruses, operator says” | <https://www.reuters.com/article/us-nuclearpower-cyber-germany-idUKKCN0XN2OS> | 26 April 2016 | Accessed 1 July 2022 | The source is publicly available information and does not contain classification markings]
- <sup>5</sup> [IEEE Xplore | Seungwon Shin, Guofei Gu, Narasimha Reddy, and Christopher P. Lee | “A Large-Scale Empirical Study of Conficker” | <https://ieeexplore.ieee.org/document/6060910> | 25 October 2011 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>6</sup> [Softpedia | Catalin Cimpanu | “Malware Shuts Down German Nuclear Power Plant on Chernobyl’s 30<sup>th</sup> Anniversary” | <https://news.softpedia.com/news/on-chernobyl-s-30th-anniversary-malware-shuts-down-german-nuclear-power-plant-503429.shtml> | 26 April 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>7</sup> [Reuters | Christoph Steitz and Eric Auchard | “German nuclear plant infected with computer viruses, operator says” | <https://www.reuters.com/article/us-nuclearpower-cyber-germany-idUKKCN0XN2OS> | 26 April 2016 | Accessed 1 July 2022 | The source is publicly available information and does not contain classification markings]
- <sup>8</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras.html) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>9</sup> [Softpedia | Catalin Cimpanu | “Malware Shuts Down German Nuclear Power Plant on Chernobyl’s 30<sup>th</sup> Anniversary” | <https://news.softpedia.com/news/on-chernobyl-s-30th-anniversary-malware-shuts-down-german-nuclear-power-plant-503429.shtml> | 26 April 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>10</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>11</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>12</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

- 
- <sup>13</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>14</sup> [SRI International | Philip Porras, Hassen Saidi, and Vinod Yegneswaran | “An Analysis of Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/index2.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/index2.html) | 4 February 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>15</sup> [SRI International | Philip Porras, Hassen Saidi, and Vinod Yegneswaran | “An Analysis of Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/index2.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/index2.html) | 4 February 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>16</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>17</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>18</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>19</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>20</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>21</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>22</sup> Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>23</sup> [SRI International | Philip Porras, Hassen Saidi, and Vinod Yegneswaran | “An Analysis of Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/index2.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/index2.html) | 4 February 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>24</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

- 
- <sup>25</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>26</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>27</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>28</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>29</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>30</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>31</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>32</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>33</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>34</sup> [Microsoft Security Intelligence | “Win32/Waledac” | <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=win32%2Fwaledac> | 13 Apr. 2009 | Accessed 02 Jun. 2022 | The source is publicly available information and does not contain classification markings]
- <sup>35</sup> [Symantec | Kelly Burton | “The Downadup Codex- A comprehensive guide to the threats mechanics” | [https://scadahacker.com/library/Documents/Cyber\\_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf](https://scadahacker.com/library/Documents/Cyber_Events/Symantec%20-%20The%20Downadup%20Codex%20v2.0.pdf) | 8 May 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>36</sup> [Symantec | Jarrad Shearer | “W32.Downadup” | <https://web.archive.org/web/20190930212315/https://www.symantec.com/security-center/writeup/2008-112203-2408-99> | 30 Sep. 2019 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>37</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]
- <sup>38</sup> [SRI International | Philip Porras, Hassen Saidi, and Vinod Yegneswaran | “An Analysis of Conficker’s Logic and Rendezvous Points” |

---

[https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/index2.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/index2.html) | 4 February 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>39</sup> [SRI International | Philip Porras, Hassen Saidi, and Vinod Yegneswaran | “An Analysis of Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/index2.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/index2.html) | 4 February 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>40</sup> [SRI International | Philip Porras, Hassen Saidi, and Vinod Yegneswaran | “An Analysis of Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/index2.html](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/index2.html) | 4 February 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>41</sup> [Microsoft | “Win32/Conficker worm” | <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=win32%2fconficker> | 7 January 2009 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>42</sup> [Usenix | “A Foray into Conficker’s Logic and Rendezvous Points” | [https://www.usenix.org/legacy/event/leet09/tech/full\\_papers/porras/porras\\_html/](https://www.usenix.org/legacy/event/leet09/tech/full_papers/porras/porras_html/) | 7 April 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>43</sup> [FireEye | “FireEye Event Description: Bot.Conficker” | <https://mil.fireeye.com/edp.php?sname=Bot.Conficker> | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>44</sup> [Microsoft Security Intelligence | “Win32/Waledac” | <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=win32%2fwaledac> | 13 Apr. 2009 | Accessed 02 Jun. 2022 | The source is publicly available information and does not contain classification markings]

<sup>45</sup> [Microsoft Security Intelligence | “Rogue:Win32/FakeSpypro” | <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Rogue:Win32/FakeSpypro> | 17 Aug. 2010 | Accessed 02 Jun. 2022 | The source is publicly available information and does not contain classification markings]

<sup>46</sup> [Microsoft Security Intelligence | “Rogue:Win32/FakeSpypro” | <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Rogue:Win32/FakeSpypro> | 17 Aug. 2010 | Accessed 02 Jun. 2022 | The source is publicly available information and does not contain classification markings]

<sup>47</sup> [Fauske & Associates | “Cost Benefit Analysis of Nuclear Power Plants” | <https://www.fauske.com/blog/cost-benefit-analysis-of-nuclear-power-plants> | 11 May 2017 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]

<sup>48</sup> [ZDNet | “Conficker estimated economic cost? \$9.1 billion” | <https://www.zdnet.com/article/confickers-estimated-economic-cost-9-1-billion/> | 23 April 2009 | Accessed 04 Jun. 2022 | The source is publicly available information and does not contain classification markings]

<sup>49</sup> [KrebsonSecurity | “\$72M Scareware Ring Used Conficker Worm” | <https://krebsonsecurity.com/2011/06/72m-scareware-ring-used-conficker-worm/> | 23 Jun. 2011 | Accessed 04 Jun. 2022 | The source is publicly available information and does not contain classification markings]

<sup>50</sup> [Softpedia | Catalin Cimpanu | “Malware Shuts Down German Nuclear Power Plant on Chernobyl’s 30th Anniversary” | <https://news.softpedia.com/news/on-chernobyl-s-30th-anniversary-malware-shuts-down-german-nuclear-power-plant-503429.shtml> | 26 April 2016 | Accessed 23 May 2022 | The source is publicly available information and does not contain classification markings]